Criterion B - Design

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Main Modules

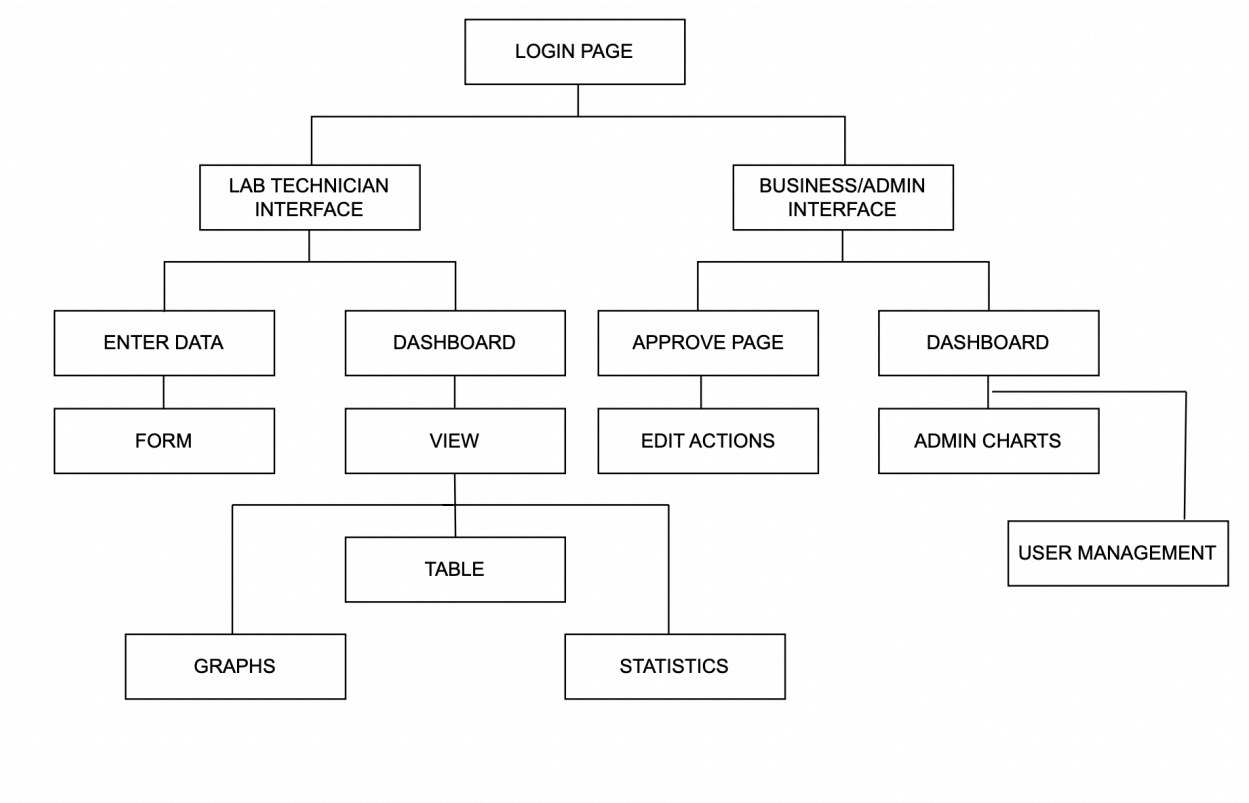


Figure 1 - Structure chart to show all modules of Chemical analysis web application

System Flowcharts

a. Login Page Processing

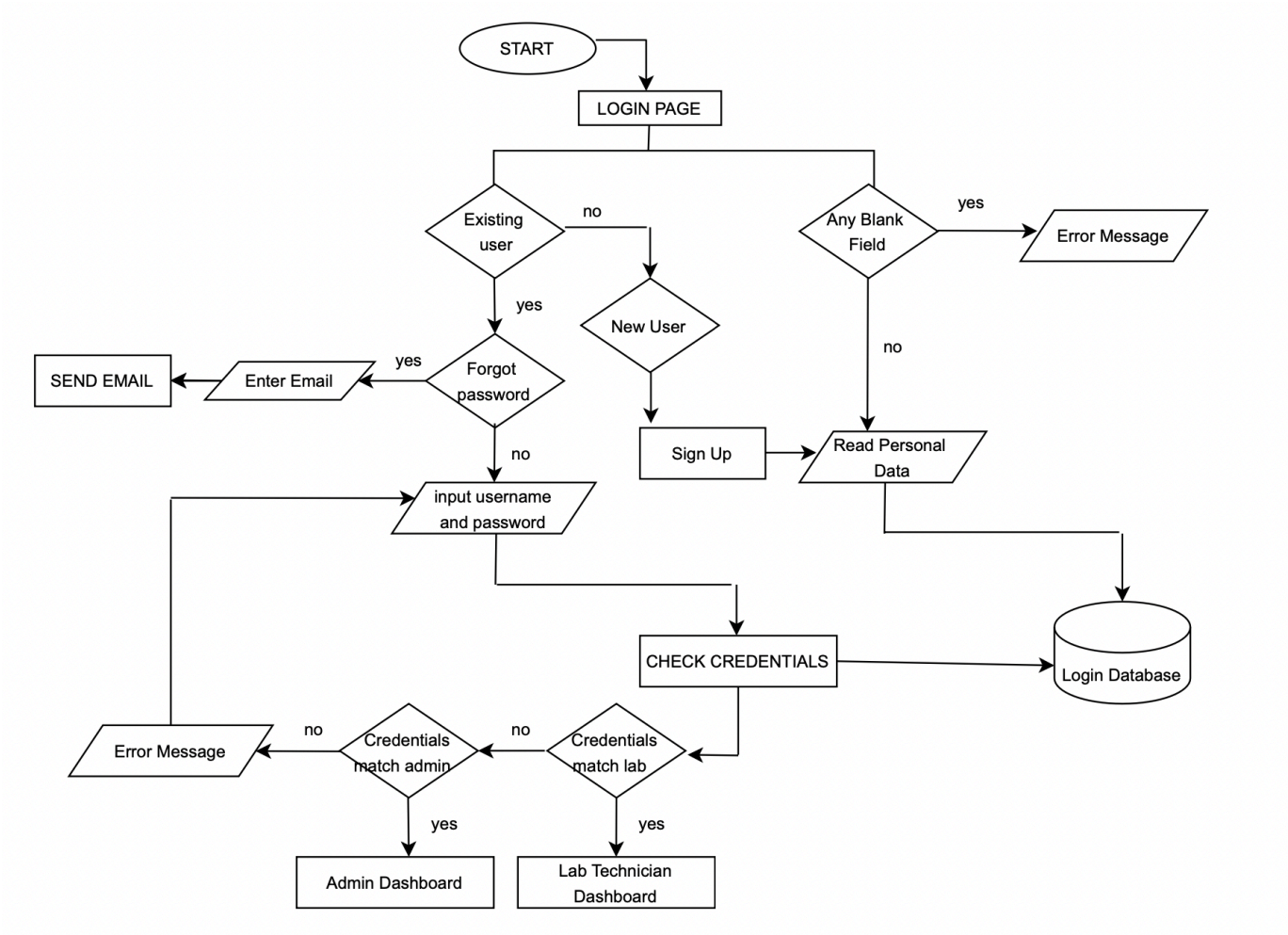


Figure 2: system chart to depict processing of Login functionality

b. System chart of the Lab Technician interface

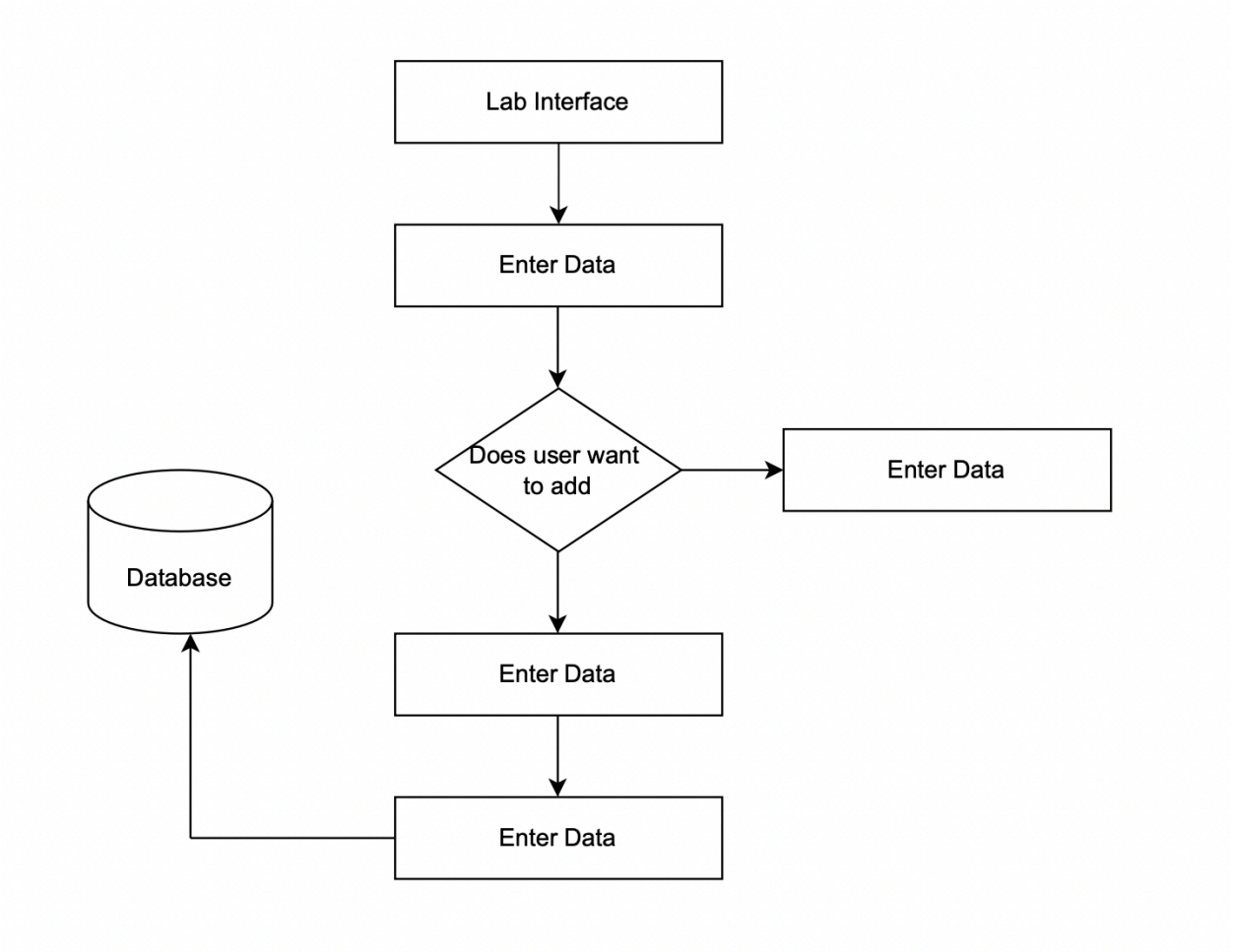


Figure 3: System chart of the Lab technician interface

c. System chart for the Lab technician to enter data

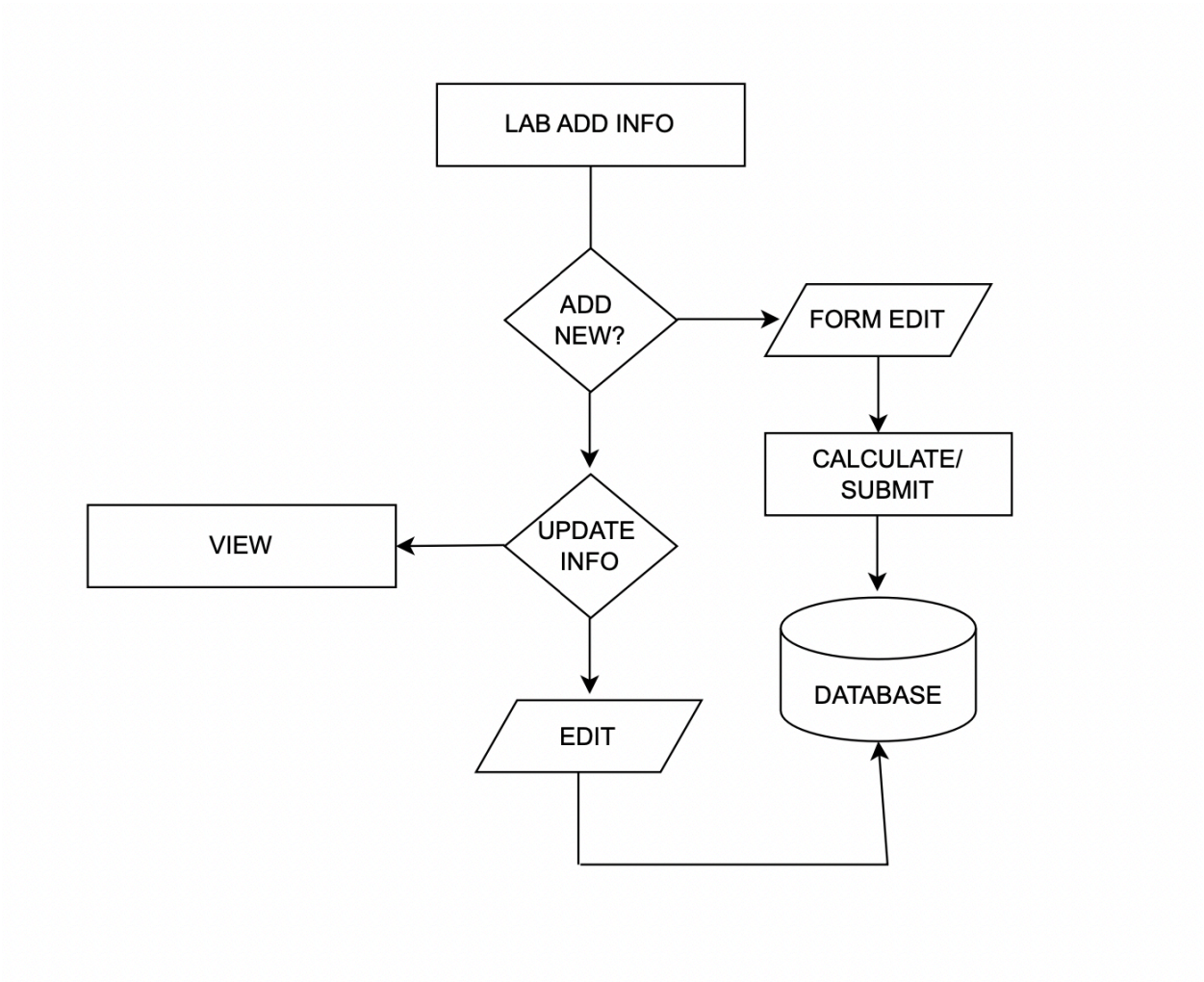


Figure 4: System chart for the Lab technician to enter data.

d. System chart for the Lab technician view

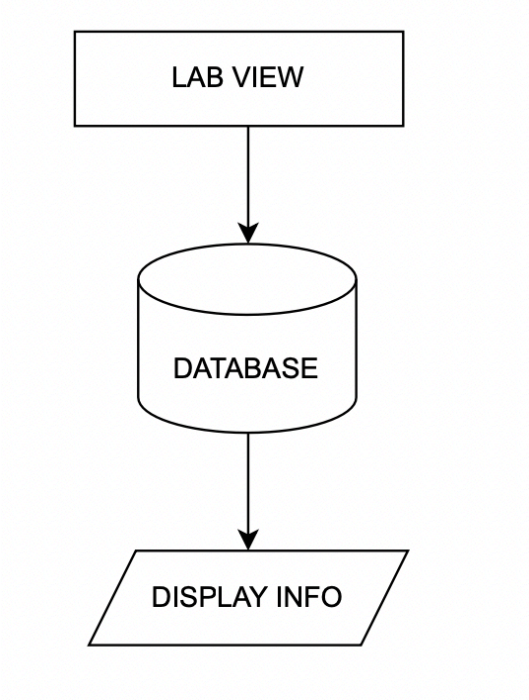


Figure 5: System chart of the Lab technician view

e. System chart of the Admin Dashboard

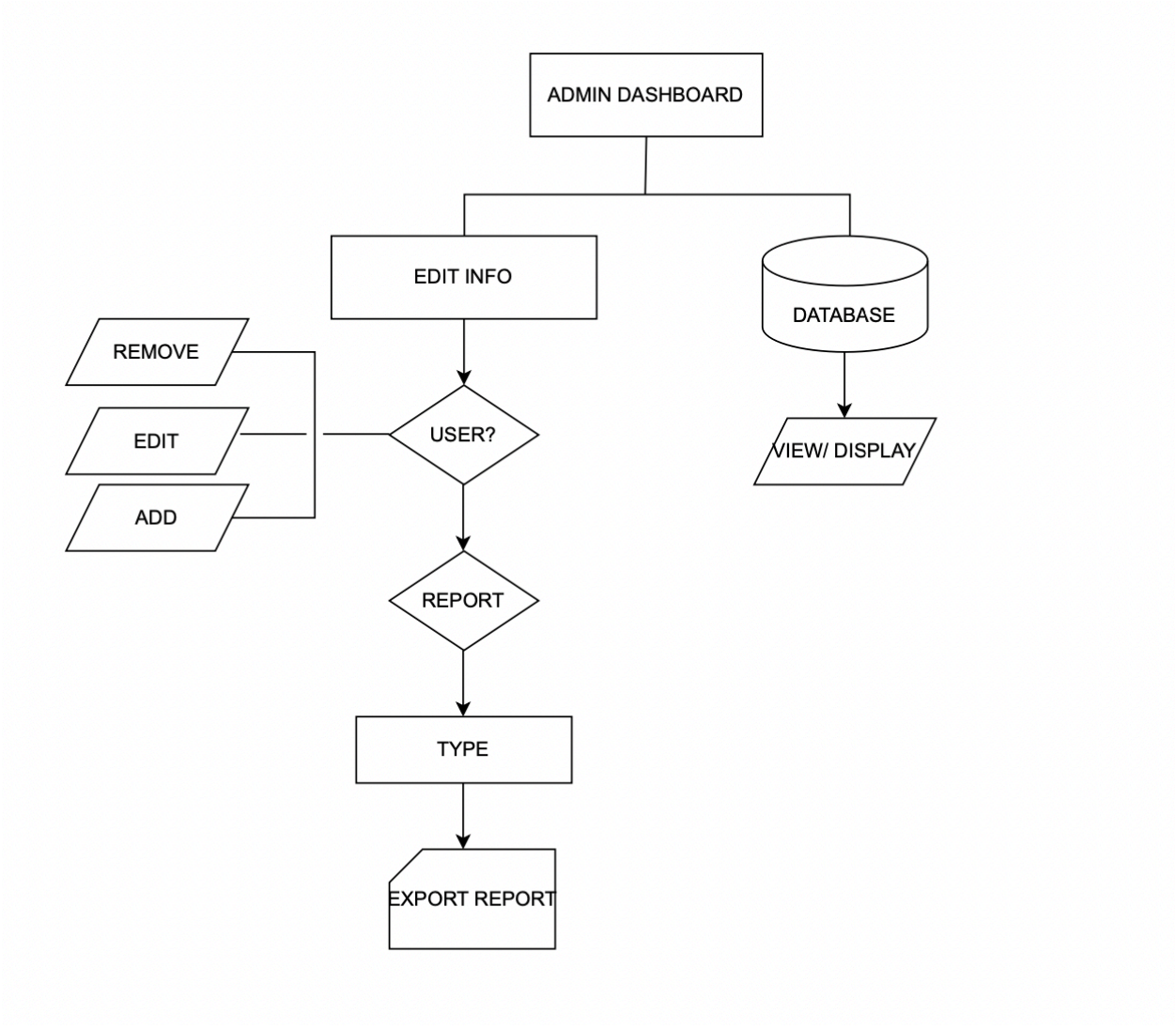


Figure 6: System chart of the Admin Dashboard

f. System chart for the Admin's Approving process

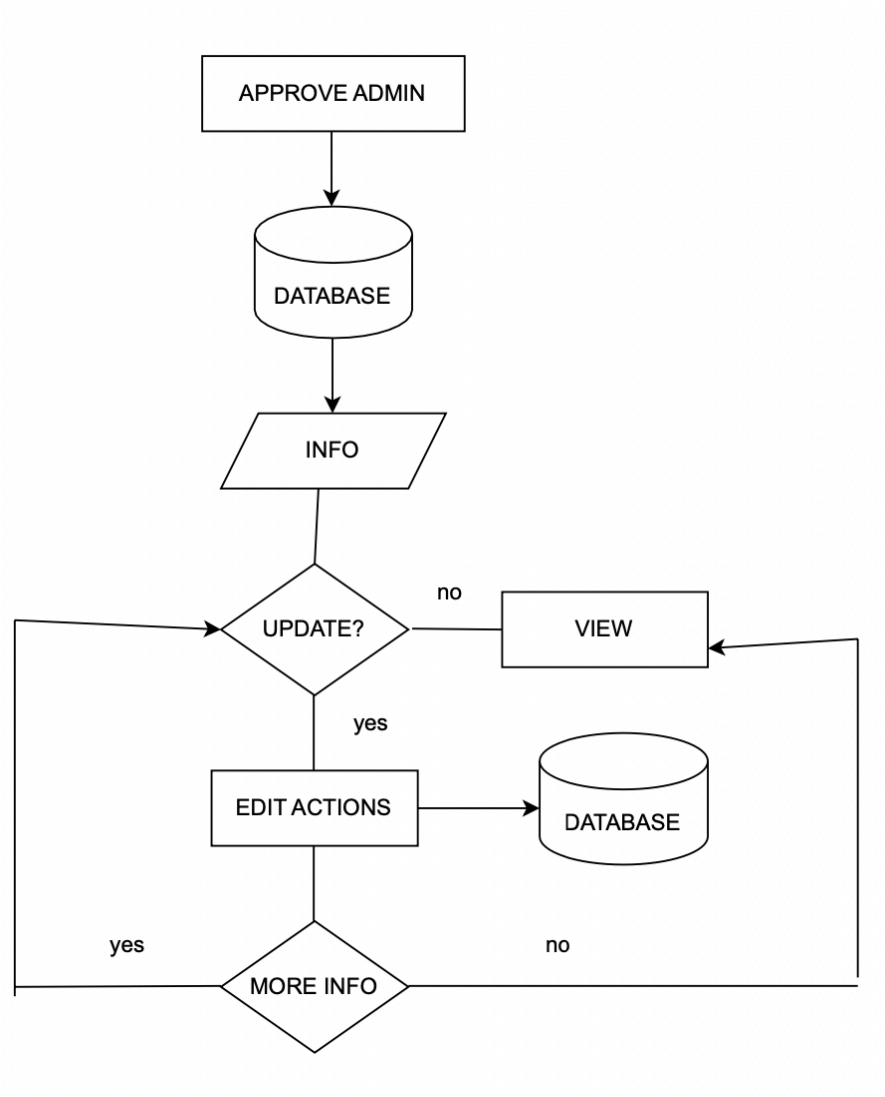


Figure 7: System chart for the Admin's Approving process

User Interface Designs

Login page

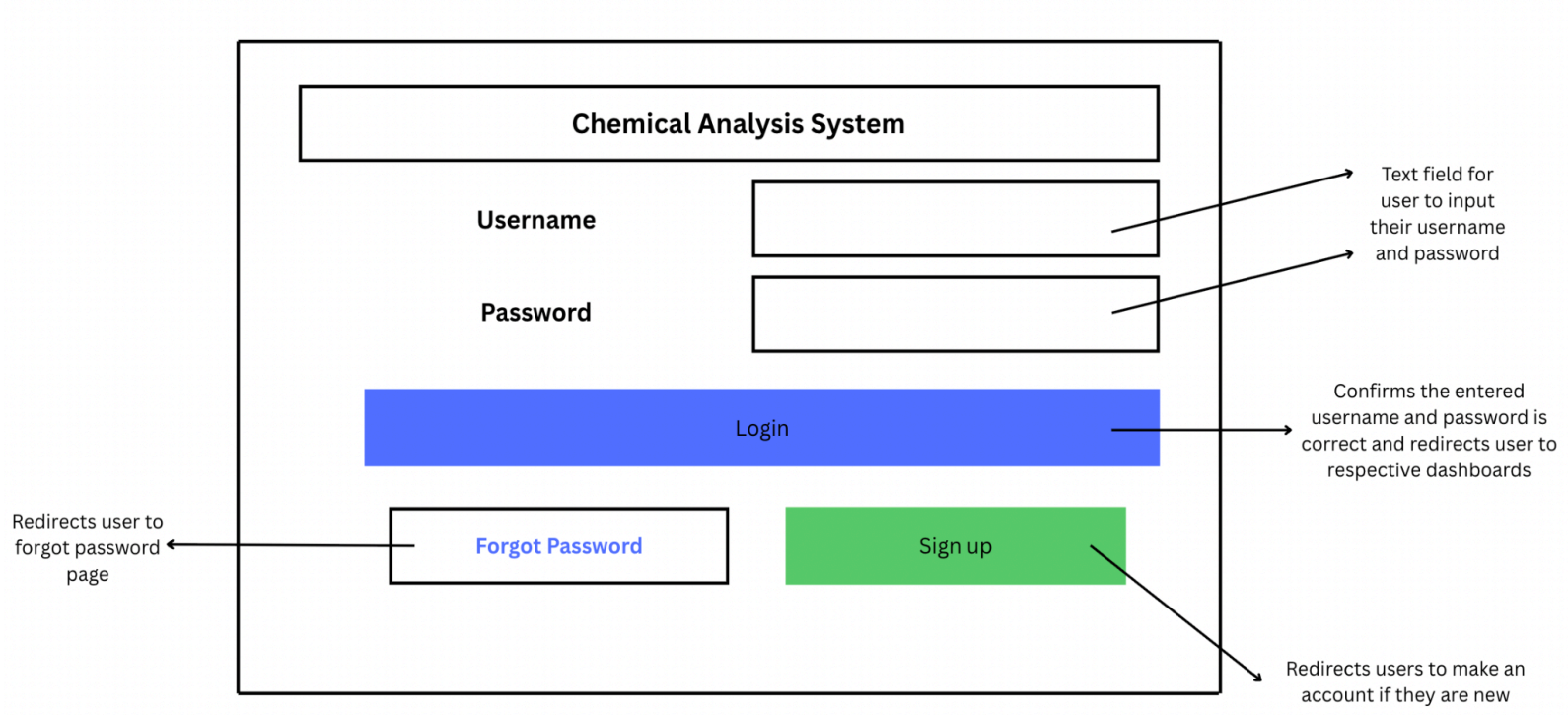


Figure 8: Graphical interface design of Login Page

Forgot password page

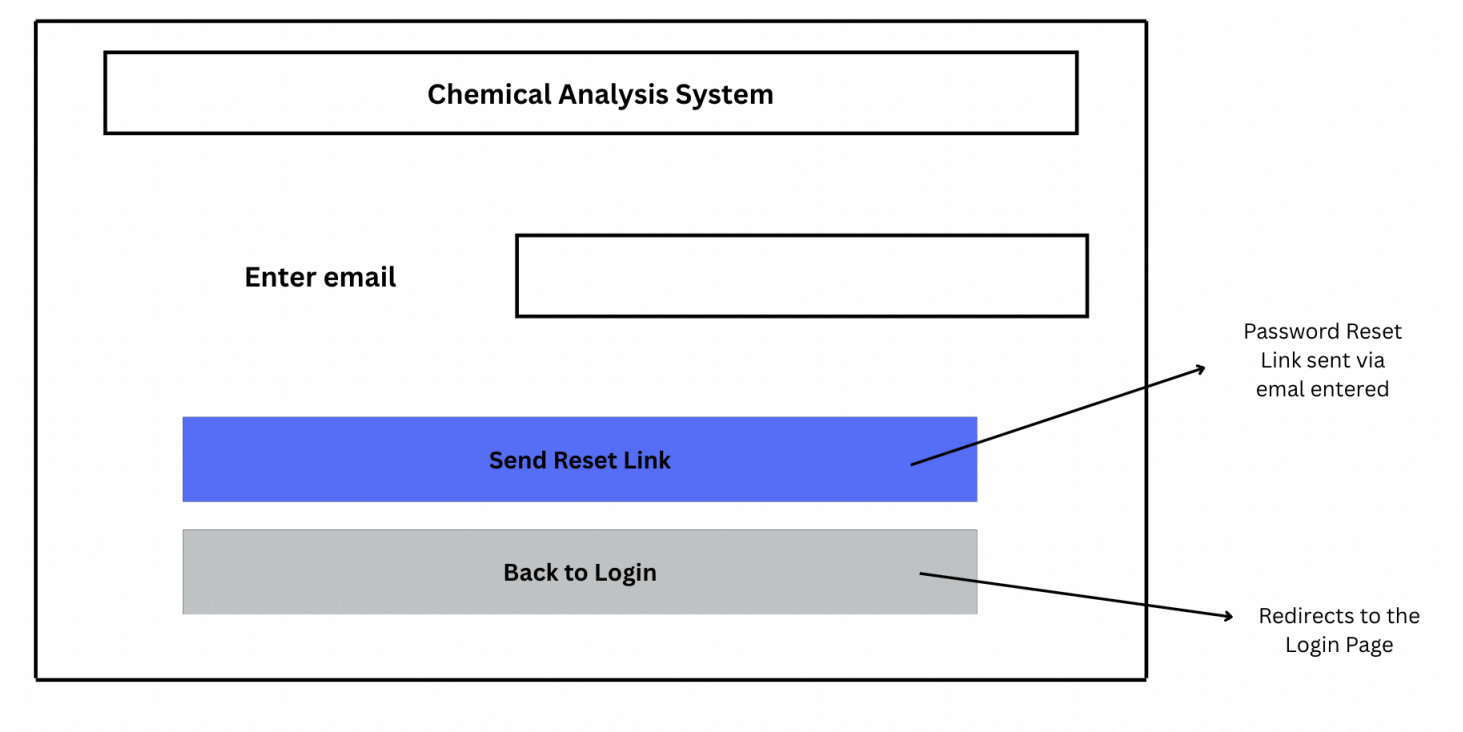


Figure 9: Graphical interface design of Forgot Password

Sign-up page

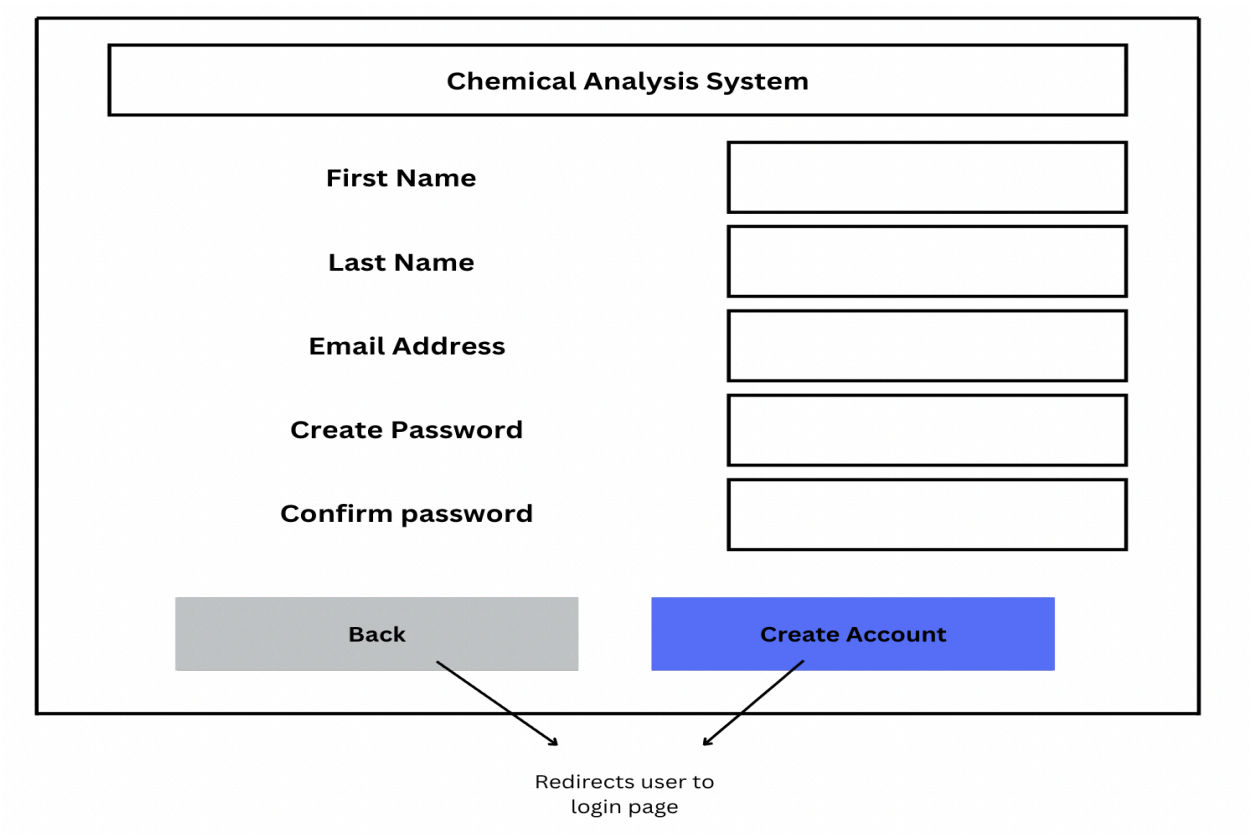


Figure 10: Graphical user interface for users to sign up

Lab technician dashboard

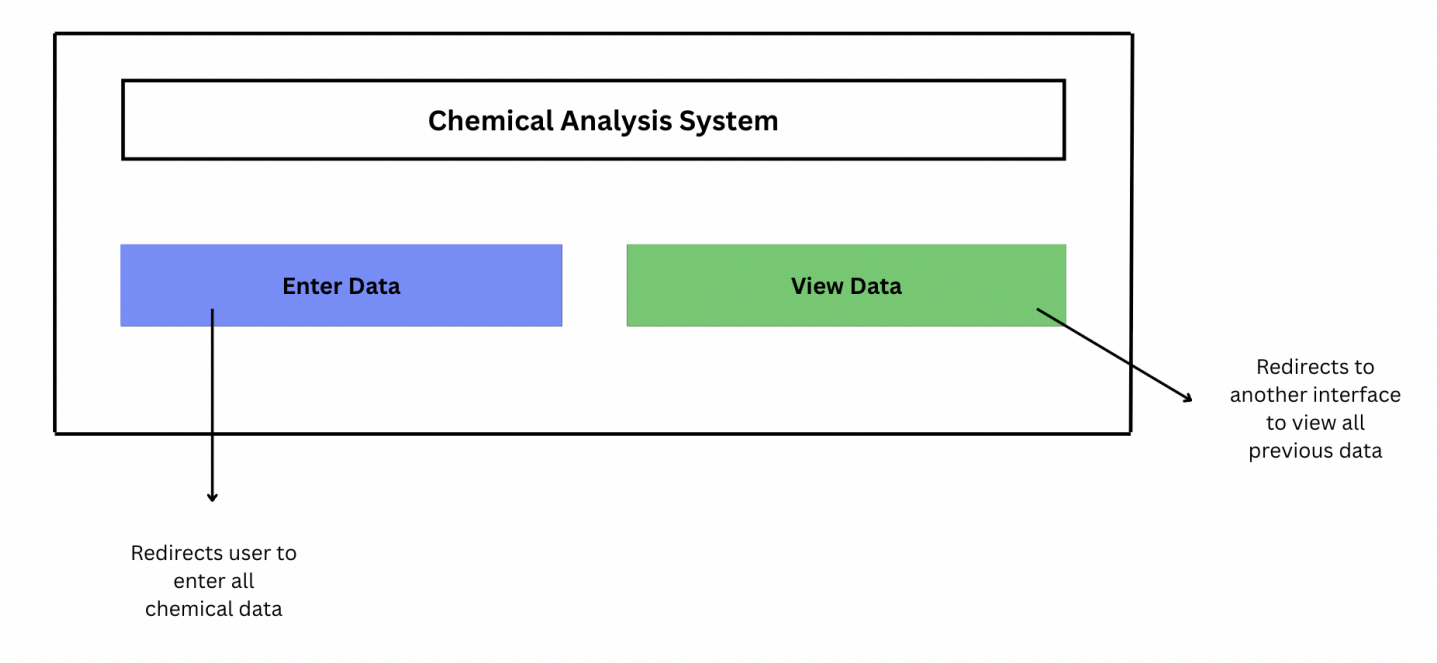


Figure 11: Graphical user interface of the lab technician dashboard

Lab technician to Enter Data Interface

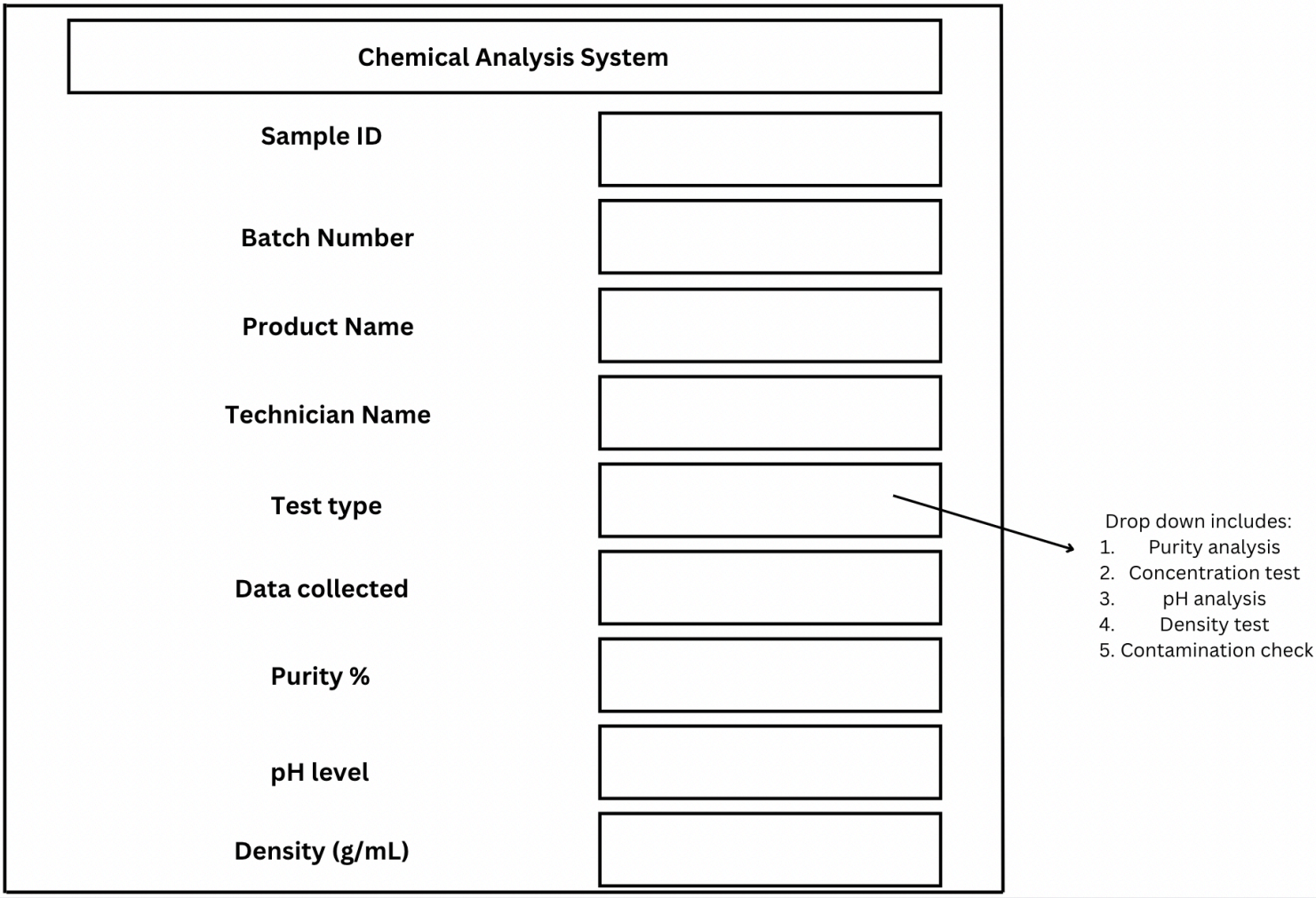


Figure 12: Graphical user interface for the Lab technician to enter data

Admin Dashboard

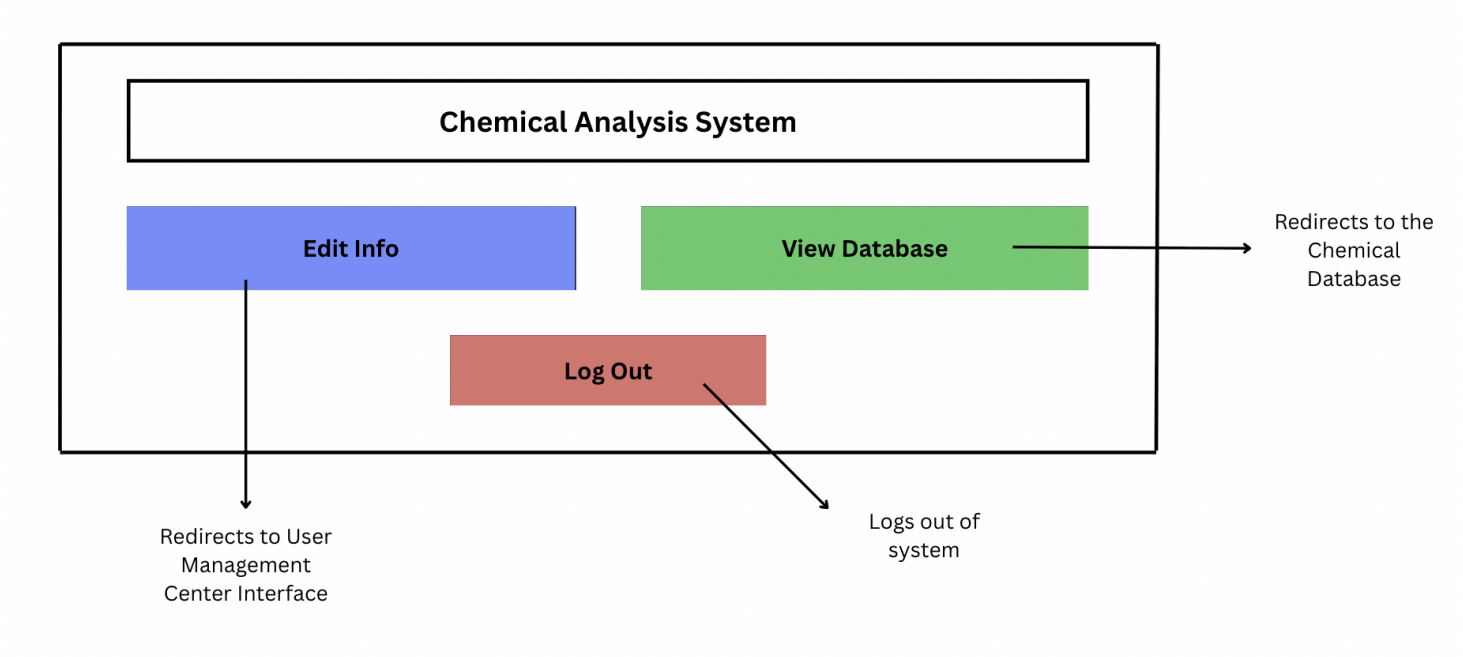


Figure 13: Graphical user interface of the Admin Dashboard

Admin interface to edit information

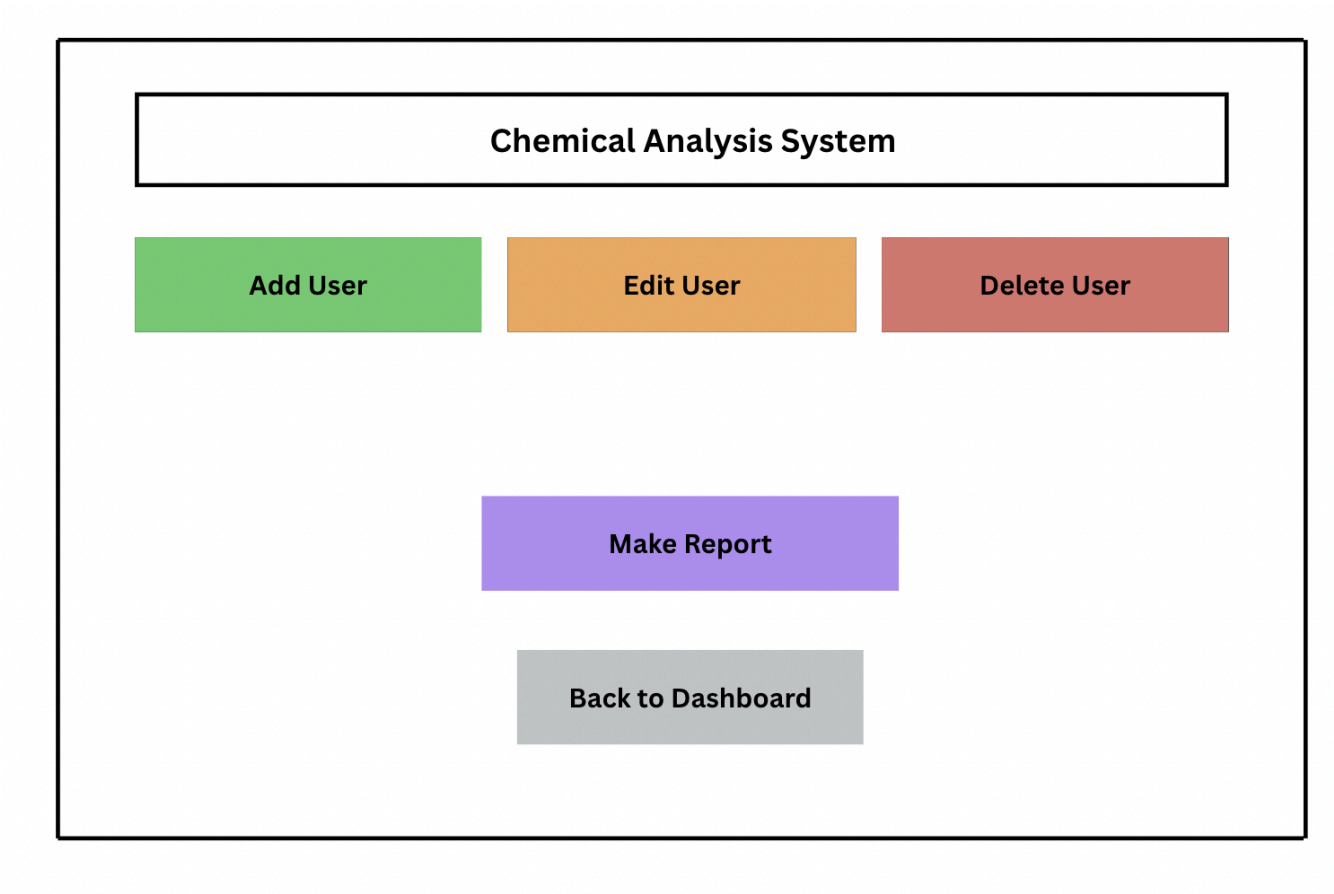


Figure 14: Graphical user interface for the Admin interface to edit information and make changes to certain users.

Report interface

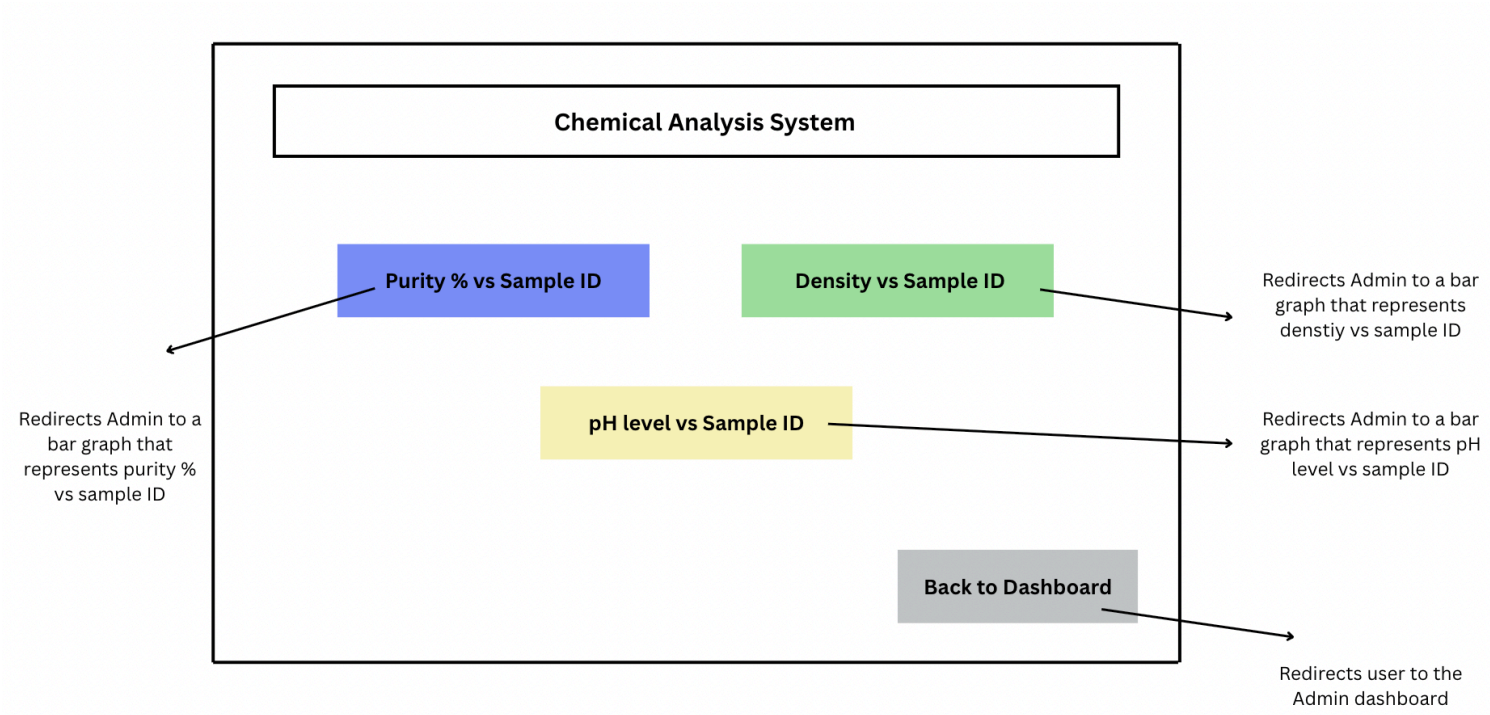


Figure 15: Graphical user interface for a choice of bar graphs

Admin bar graph 1

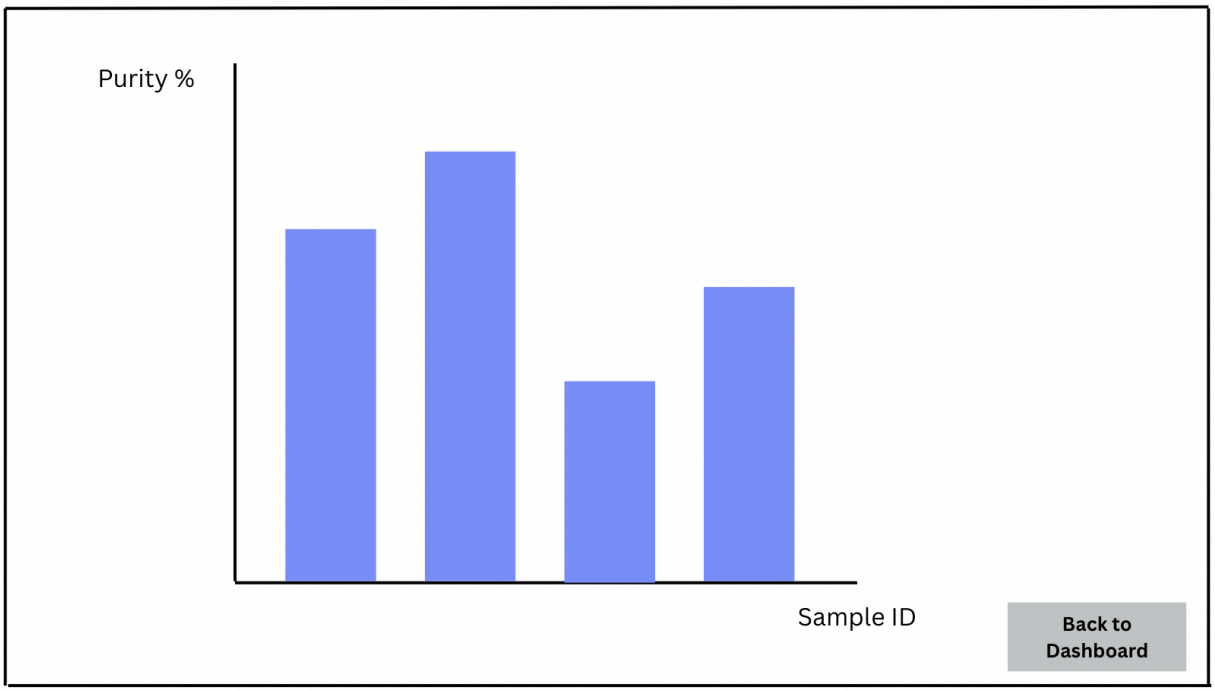


Figure 16: Graphical user interface that represents Purity % and Sample ID

Admin bar graph 2

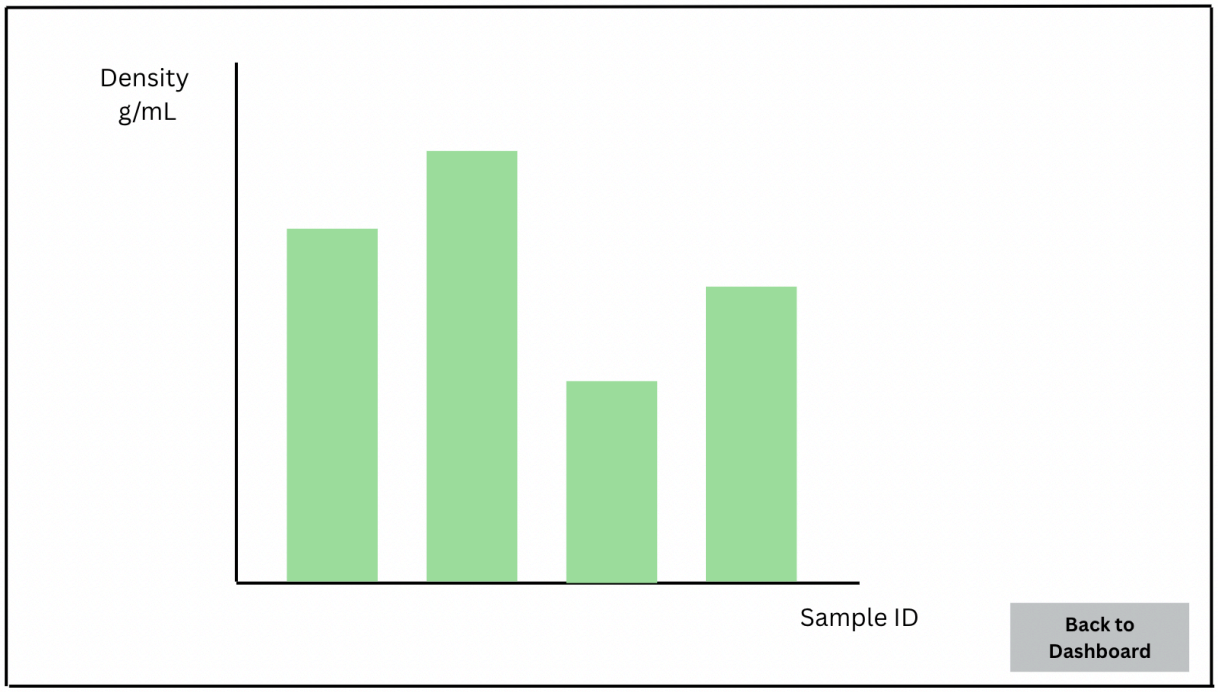


Figure 17: Graphical user interface that represents Density g/mL and Sample ID

Admin bar graph 3

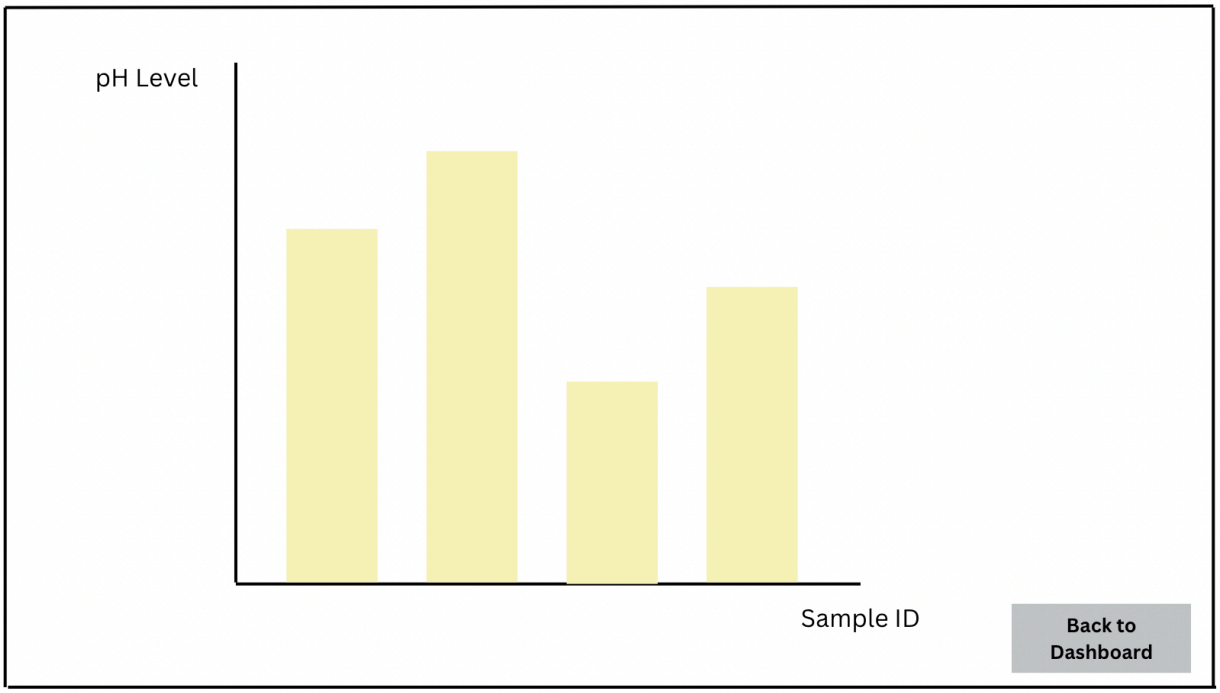
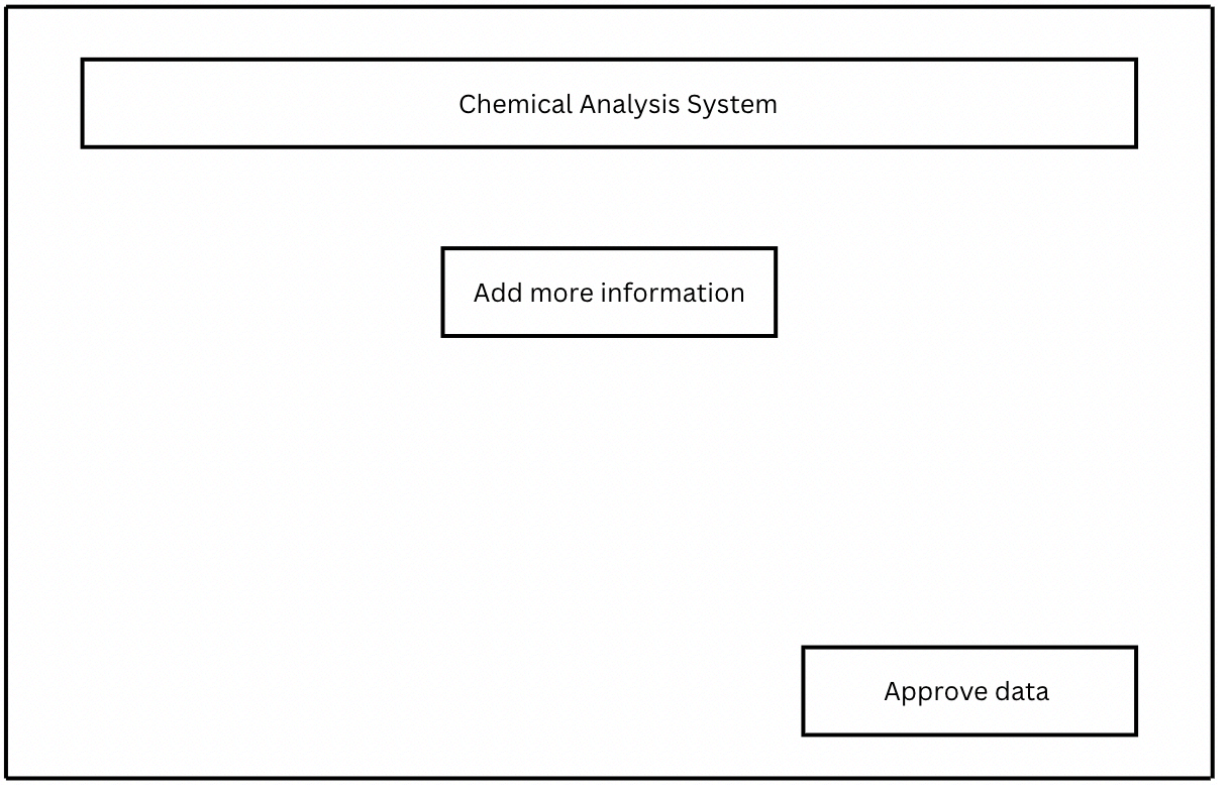


Figure 18: Graphical user interface that represents pH level and Sample ID

Admin interface to view and approve data



Database Tables

Login_tbl		
Field	**Datatype**	**Purpose**
Email Address (Primary Key)	String	To hold the unique ID of the user
First Name	Text / String	To store the first name
Last Name	Text / String	To store the last name
Category	Text/ String	Store “Admin” or “Laboratory Technician”

Table 1: Login Database

The database contains all the required credentials for accessing the respective portals (Admin or Laboratory Technician). A user is granted access only if the entered details match the stored records.

Lab_technician_details_tbl		
Field	**Datatype**	**Purpose**
Username (Primary Key)	Text / String	To store the unique username of the lab technician
First Name	Text / String	To store the last name
Last Name	Text / String	To store the password

Table 2: Laboratory Technician Database

The database contains the necessary details about the Laboratory technician details that are entered when they sign up. The web application identifies these details and redirects them to the respective dashboard

Admin_Details_tbl		
Field	**Datatype**	**Purpose**
Username (Primary Key)	Text / String	To store the unique username of the lab technician
First Name	Text / String	To store the last name
Last Name	Text / String	To store the last name

Table 3: Admin Details table

The database contains the necessary details about the Admin details that are entered when they sign up. The web application identifies these details and redirects them to the respective dashboard

Chemical_Details_tbl		
Field	**Datatype**	**Purpose**
Sample ID (Primary Key)	Text / String	Unique identifier assigned to each chemical sample.
Batch Number	Text / String	Stores the batch number associated with the sample
Product Name	Text / String	Stores the product name
Technician ID	Text / String	Identifier for the technician conducting the test
Test Type	Text / String	Indicates the type of test performed on the sample
Purity %	Number	Records the percentage purity of the sample
pH Level	Number	Stores the pH value of the sample
Density	Number	Captures the density

		measurement of the sample

Table 4: Chemical sample table

This database structure provides a comprehensive and organized way to manage chemical testing data

Test_results_tbl		
Field	**Datatype**	**Purpose**
Result ID (Primary Key)	Text / String	Uniquely identify each individual test result record
Sample ID (Foreign Key)	Text / String	Link each test result to a specific chemical sample
Technician ID (Foreign Key)	Text / String	Identify which lab technician performed and recorded this specific test result
Test Date	DD/MM/YY	Record the exact date when the chemical analysis or test was conducted
Test type	Text / String	Specify the kind of analysis or test that was performed
Remarks	Text / String	Any additional notes, observations, or comments related to the specific test result.

Table 5: Test results table

Approvals_tbl		
Field	**Datatype**	**Purpose**
Approval ID (Primary Key)	Text / String	uniquely identify each individual approval record
Sample ID (Foreign Key)	Text / String	link each approval record to the specific chemical sample that is being approved or reviewed.
Admin ID	Text / String	identify which administrator performed the approval action
Status	Text / String	indicate the current state or outcome of the approval process for a given sample (approved, pending or rejected)

Table 6: Approvals Table

Functionalities of each class

Class Name	Functionality
Login Page	This contains the code to retrieve the username and password from the Login_tbl database table and on the basis of matched category (admin or lab technician) redirect the page to their respective account portal.
Sign Up Page	Allows signing up for new lab technicians for the first time by asking for information such as first name, last name, their email address and making a password to complete the process.
Forgot Password Page	This is common to all users, it contains a code which retrieves the password of the user in case the user forgets their password. This is done by sending a reset password link to the email address that the user enters.
Laboratory Technician Dashboard	This contains the code to redirect the student user to different avenues such as enter data and view data.
Enter Chemical Data	Allows the lab technician to enter chemical data into the system that includes, sample ID, batch number, product name, technician name, test type, purity %, pH level and the density of the sample.
View Chemical Data	Allows the lab technician to view all the chemical data that has previously been entered.
Admin Dashboard	This contains the code to redirect the student user to different avenues such as view database and edit user management information.
View Database	Allows the admin to view the database and all information that has been added either by them or by the lab technicians previously
Add User	Allows the admin to create new user accounts with custom permissions and access levels for the chemical analysis system
Edit User	Allows the admin to modify existing user profiles, update their permissions and manage account settings and preferences.
Delete User	Allows the admin to safely deactivate or permanently delete

	user accounts while preserving important data records.
Make Report	Allows the admin to make reports of all the chemical data and gives the admin 3 choices for graphs which are 1 - purity % vs Sample ID. 2 - Density vs Sample ID. 3 - pH level vs Sample ID
Approve Chemical Data	Allows the admin to approve the chemical data that is added by the laboratory technicians, ensuring that they are correct

Test Plan

Test No.	Test Type	Nature/ Methodology of Test	Input Data	Expected Outcome
1.	The users should be able to login into the respective portal successfully using respective credentials such as Admin dashboard and Lab technician dashboard.	Logging in with different types credentials available in database table login_tbl	**Normal data set**: Correct username and password. **Abnormal data set**: Correct username and Wrong password, Wrong username and Correct password, Wrong username and password	Based on correct credentials and different values in category field of database; page will redirect to the respective portal
2.	A registered user should be able to receive a new password through email if the password is forgotten.	Click on the forgot password button. To verify a user's identity, users need to correctly enter their email address chosen when creating an account.	An appropriate email ID should be entered that contains '@' and '.com', and it must be registered in the database table	The user will receive an email containing the new password. After re-entering this password, the software will validate it along with the registered username. If there is a network connection failure, an appropriate error message will be displayed.

3.	The system will allow lab technicians to sign up and create a new account with a unique email ID and password	Click on the “Sign up” Button	**Normal Data**: unique email ID and password. **Abnormal Data**: No student can enter details that are already present in the login_tbl in the database	Lab technicians and admins will be able to successfully create their respective new accounts with their unique email IDs and password. It will then redirect the user back to the login page and thereafter their respective dashboard
4.	A lab technician should be able to enter chemical data into the database/system	Click on the “Enter Data” Button	**Normal data set:** Correct chemical name and valid numerical values. **Abnormal data set:** Correct chemical name and invalid numerical values, Wrong chemical name and valid numerical values, Wrong chemical name and invalid numerical values.	Lab technicians will be redirected to another interface where they will be able to view all the previous data that has been added in the database
5.	The admin will be redirected to the user management center interface.	Click on the “Edit Info” button		The admin will be redirected to another interface where they can edit information about already existing users. They can add and delete users

				as well.
6.	The admin will be redirected to the chemical database	Click on the “View Database” button		The admin will be redirected to the chemical database where they will be able to view all the chemical information.
7.	The system should allow the admin to generate reports using bar graphs.	This test is carried out by selecting the “Make Report” option and choosing one of the available graph types: ● Purity % vs Sample ID ● Density vs Sample ID ● pH level vs Sample ID		If valid data exists, the selected bar graph is generated accurately. If insufficient data exists, the system displays a message indicating that the report cannot be generated.
8.	The system should allow the admin to review and approve chemical data submitted by lab technicians.	This test is conducted through the Approve Chemical Data interface.		If the data is complete and accurate, the admin can approve it and the approval status is updated in the database. If the data is

				incomplete, the system allows the admin to request additional information instead of approving.
9.	The system should allow lab technicians to view previously entered chemical data.	This test is performed by clicking “View Data” from the Lab Technician dashboard.		If records exist, the system displays all relevant chemical data clearly. If no records exist, a message is shown stating that no data is available.
10.	The system should restrict access to features based on user roles.	This test is conducted by attempting to access restricted pages using incorrect roles.		Users can only access features assigned to their role. Unauthorized access attempts are blocked, and the user is redirected or shown an error message.

Test type, nature, input, expected outcome